

CALEB MANTEY

XR / SPATIAL COMPUTING &
GAME ENGINEER

CONTACT

Accra, Ghana

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CORE SKILLS

XR & real-time 3D

Unity, C#, WebXR, Three.js, React
Three Fiber, OpenXR, Meta Quest,
VR / AR / MR

Spatial systems

3D interaction, simulation, multiplayer
networking, digital twins, GIS &
spatial data

Engineering

TypeScript, React, Node.js, Blender,
WebRTC, performance optimization

Leadership

Technical direction, team leadership,
product delivery, public speaking

EDUCATION

B.Sc., Geomatic Engineering

University of Mines and Technology
(UMaT), Ghana

Focus: spatial data, GIS, coordinate
systems

RECOGNITION

- 2023 NEIP Presidential Pitch, overall winner (Relu Interactives).
- 2023 Ghana Innovation & Startup Awards, Animation/Gaming Startup Innovation of the Year.

LINKS

Portfolio

GitHub

LinkedIn

YouTube

itch.io

OPEN SOURCE & TEACHING

Publish a Unity tutorial series on
YouTube (@CalebMantey) that builds
a third-person bow and arrow system.
Playable demos on itch.io.

PROFILE

XR & game developer, Co-Founder/CTO of Relu Interactives, and lead engineer of Relu Spatial. I have pioneered immersive technology in Africa since 2016, building VR training simulators, industrial digital twins, multiplayer 3D, and Unity games, plus the browser-native WebXR engine behind Relu Spatial. I studied Geomatic Engineering, so I bring training in spatial data and coordinate systems, the same math XR runs on.

EXPERIENCE

Relu Interactives · Co-Founder & CTO

Sep 2022 – Present

- Built Relu Spatial end-to-end, from backend services to the real-time XR engine, enabling AI-assisted 3D creation and cross-device publishing to web, mobile, and headsets.
- Directed PetSim, a petroleum drilling-rig training simulation endorsed by five Ghanaian universities and offered free to students; it anchored Relu's 2023 NEIP Presidential Pitch win.
- Delivered VIVaTS, an operator-training and live-monitoring digital twin of national gas infrastructure for Ghana National Gas Company.
- Shipped 10+ immersive products across training, digital twins, brand activations, and VR games on Unity/C#, WebXR, and Meta Quest/OpenXR.
- Lead cross-disciplinary teams (engineering, 3D art, design) from prototype to delivery; represent the company in industry talks and workshops.

Andela / client projects · Software Engineer, Web 3D & Games

Sep 2021 – Jan 2024

- Built and ported browser-based 3D games and WebXR prototypes for distributed client teams; integrated gameplay with backend services and fixed production performance issues.

Soko Aerial Signal Training School · AR / VR Developer & Researcher

Sep 2019 – Aug 2020

- Prototyped Unity VR/AR experiences for military training and spatial awareness; mentored junior developers.

SELECTED PROJECTS

Relu Spatial · Lead engineer; backend + XR engine for an AI-assisted 3D/WebXR creation and publishing platform.

PetSim · Petroleum drilling-rig training sim, free to every student in Ghana and endorsed by five universities.

VIVaTS · Operator-training and live-monitoring digital twin for Ghana National Gas Company.

Africa's Legends VR · VR adaptation of African superhero stories, built with Leti Arts.

SortJam VR · Physics-based VR sorting game with timed challenges and themed-world progression.

Dark Echoes + itch.io · Unity third-person shooter and other playable demos published on itch.io.